Using AGs to migrate SQL Server with little downtime

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https://github.com/InfoJunkie1/AG4Migrations

St. Louis SQL Server and Power BI User Group

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About Me

- High school and then college writing instructor
- Needed full-time job!
- Went through LaunchCode SQL cohort
- Hired at NISA!
- Asked to come back to LaunchCode as a TA and then asked to be a lead mentor
- Passionate about database security
- Honor to have presented at PASS with my daughter!!!

(Giza—Great Pyramid)

Expectations

- You may have heard of Availabiity Groups, but you don’t need to know much about them
- If you use AGs currently, you would use distributed AGs for migration
  - Out of scope
- Familiar with SQL Server
- You are interested in exploring options to expedite migrating SQL Server to a new version

(Giza—Great Pyramid)

30 seconds on AGs

- WSFC on the OS level, not SQL Server
- WSFC needs quorum
  - Don’t want a tie!
  - Create File Share Witness as tie breaker
- AGs inside SQL Server
- AG ensures databases on both servers are synchronized
- One instance is called Primary
  - SQL19
- Other instance is Secondary
  - SQ25

(Giza—Sphinx)

Advantages of migrating via AGs

- Little down time
- Ability to shift the time-intensive part to before cutover day
- Did I mention very little down time?

(Valley of the Kings—Tomb of Tausert/Setnakht Tausert)

Items to watch out for

- AG, WSFC temporary state
- Mixed versions Microsoft approved for short term
- Only go up one version of Windows OS
- Watch storage during seeding
- Compat levels preserved
  - Update if tested
- SQL Server 2025 breaking changes with encrypted connections

(Valley of the Kings—Tomb of Tutankhamun)

Windows Server Failover Cluster

- May need to coordinate with sysadmins, storage admins
- Set up on each migration pair
- Set up on current server
- Make sure you set up File Share Witness

- Demo

(Beni Hassan—Tomb of Khnumhotep II)

Create an AG

- GUI or script
- Requires a SQL Server service restart
- Copy logins
  - Use DBAtools to include SIDS, hashed passwords
- Set up new sql server with all other server-level configurations
  - Jobs that reference database need to be copied over after
- System databases will be new so update as needed

(Giza—Mena House hotel balcony)

Seeding methods

- Requirements
  - Full
  - Backup taken
- Automatic seeding (easiest!)
  - Watch transaction log
  - Demo
- Join only
  - Uses tail-of-the-log backup
  - VLDB
- Hybrid
  - Best for VLDB
  - No need for tail-of-the-log!!!!!
  - Demo

Luxor—Luxor Temple

Manual failover

- One way trip
- Update from async commit to sync commit
- Failover!

(Nile cruise)

Clean up

- Drop AG
- Database Master Key relinked to new Server Master Key
- Leave databases on original server in Restoring

(Petra—Treasury)

Rollback

- Take databases out of AG if still there
- Restore with recovery

(Petra—Monastery)

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(Cairo—tattoo)